

SATYA BHARGAV MADDIPATI

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Education

University of Wisconsin–Madison

Madison, WI

M.S. in Computer Sciences — GPA: 4.0/4.0

Sep 2025 – May 2027 (Expected)

- **Coursework:** Software Engineering, Database Systems, Big Data, Computer Vision, Deep Learning, AI Agents

Indian Institute of Technology Bombay

Mumbai, India

B.Tech-M.Tech Dual Degree with Honors — GPA: 8.9/10

Aug 2018 – Jun 2023

Experience

Tata AIA Life Insurance | Software Engineer – Machine Learning

Mumbai, India

• **PYTHON** • **FASTAPI** • **REST APIS** • **MICROSERVICES** • **DOCKER** • **KUBERNETES** • **AZURE** • **CI/CD** *Jun 2023 – Aug 2025*

- **Designed and deployed scalable FastAPI microservices** backing a customer support platform serving **50K+** users on Azure, sustaining **0.8s p95 latency** and **99.5% uptime** under production load
- **Owned design through delivery** of production workflow services integrating **LangGraph/CrewAI** agents into existing claims systems, automating multi-step processing and cutting manual review effort by **35%**
- Built a **reusable, configurable training pipeline** that fine-tuned **5+ open-source LLMs (Llama 3.2, Qwen 2.5)** via **QLoRA/PEFT**, shipped with **automated tests and CI/CD**, validated through peer code review, and reached **94%** accuracy on internal benchmarks
- Designed a modular recommendation service over **2M+** customer records using embeddings and collaborative filtering, exposed as a **REST API** consumed by multiple internal teams

IIT Bombay | Graduate Research Assistant

Mumbai, India

• **PYTHON** • **DATA PIPELINES** • **PANDAS** • **NUMPY** • **TIME SERIES** *Apr 2022 – Jun 2023*

- Built a modular forecasting pipeline for air-quality data: stochastic **KZ filtering** to strip meteorological trend, then swappable ensemble models (**XGBoost, Random Forests, SARIMA**) behind a common interface
- Ran **SHAP-based interpretability analysis** to quantify seasonal and feature-level contributions in the forecasts

Cognibot AI | Machine Learning Intern

Mumbai, India

• **COMPUTER VISION** • **OBJECT DETECTION** • **EFFICIENTDET** • **PYTHON** *Dec 2021 – Feb 2022*

- Built a scalable people-tracking system for retail stores on an **EfficientDet** detector, profiling and tuning it across varied **compute and memory constraints** to meet real-time throughput targets

Projects

Reason-Guessr: Image Geolocation | • **CLIP ViT-B/32**

reason-guessr.vercel.app

- Built and **deployed an interactive web application** that predicts GPS coordinates from a single street-level image, wiring a browser frontend to a Python inference service and shipping it to production on **Vercel**
- Extracted interpretable cues (climate, plates, vegetation, architecture) zero-shot via a **CLIP ViT-B/32** pipeline and fused them with visual embeddings for coordinate regression; on a self-built **32K-image** dataset over **8,000** locations, cue fusion improved median haversine error from **90.6 km** to **88.9 km**

Teacher-Student Retrieval Learning | • **QWEN3** • **RETRIEVAL** • **DISTILLATION**

In progress

- Training a **Qwen3-1.7B** student against a frozen **4B teacher** to learn efficient when-to-search / what-to-retrieve behavior under explicit **tool-call costs**
- Evaluating on controlled fact-finding tasks and **MuSiQue** multi-hop QA, measuring answer accuracy, retrieval cost, and abstention, against standard knowledge distillation and information-gain search baselines

Grounded Vision-Language Agents for Instruction Following | • **LLAVA-1.6** • **LORA/PEFT**

- Built a multimodal agent on **LLaVA-1.6 (7B)** with a custom **Observe→Reason→Act** loop for multi-step instruction following on **VisualWebBench** and **ScienceQA**, benchmarked against text-only **Mistral-7B + ReAct** baselines

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, R

Front-end: React, Next.js, Vue.js, React Native, Node/Express.js, HTML/CSS

Back-end & Data: FastAPI, Spring Boot, REST APIs, PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, LangChain, LangGraph, CrewAI, QLoRA/PEFT, Pandas, NumPy, SHAP, OpenAI/Anthropic/Gemini APIs

Infrastructure: AWS, Azure, Docker, Kubernetes, Kafka, RabbitMQ, Prometheus, Grafana, CI/CD, Linux

Core CS & Systems: Data Structures & Algorithms, Operating Systems, Computer Networks, Distributed Systems, Microservices, Object-Oriented Design

Achievements and Positions of Responsibility

Achievements: JEE Advanced Top 0.5% among 250K candidates; **Finalist**, Tata Innovation Workshop (**25+** teams) — WhatsApp AI cross-sell flow cutting add-on processing from **15 days to under 5 min**; Inter-IIT Sports Bronze Medal

Positions of Responsibility: Web Coordinator, Techfest IIT Bombay